A First Course in PROBABILITY

EIGHTH EDITION



SHELDON ROSS

$$
\begin{aligned}
P\{R \le a\} &= P\{X_{(n)} - X_{(1)} \le a\} \\
&= \iint\limits_{x_n - x_1 \le a} f_{X_{(1)},X_{(n)}}(x_1, x_n)\, dx_1\, dx_n \\
&= \int_{-\infty}^{\infty} \int_{x_1}^{x_1+a} \frac{n!}{(n-2)!}[F(x_n) - F(x_1)]^{n-2} f(x_1) f(x_n)\, dx_n\, dx_1
\end{aligned}
$$

Making the change of variable $y = F(x_n) - F(x_1), dy = f(x_n)\, dx_n$ yields

$$
\begin{aligned}
\int_{x_1}^{x_1+a} [F(x_n) - F(x_1)]^{n-2} f(x_n)\, dx_n &= \int_0^{F(x_1+a)-F(x_1)} y^{n-2} dy \\
&= \frac{1}{n-1}[F(x_1 + a) - F(x_1)]^{n-1}
\end{aligned}
$$

Thus,

$$
P\{R \le a\} = n \int_{-\infty}^{\infty} [F(x_1 + a) - F(x_1)]^{n-1} f(x_1)\, dx_1 \tag{6.7}
$$

Equation (6.7) can be evaluated explicitly only in a few special cases. One such case is when the X_i's are all uniformly distributed on (0, 1). In this case, we obtain, from Equation (6.7), that for $0 < a < 1$,

$$
\begin{aligned}
P\{R < a\} &= n \int_0^1 [F(x_1 + a) - F(x_1)]^{n-1} f(x_1)\, dx_1 \\
&= n \int_0^{1-a} a^{n-1}\, dx_1 + n \int_{1-a}^1 (1 - x_1)^{n-1}\, dx_1 \\
&= n(1 - a)a^{n-1} + a^n
\end{aligned}
$$

Differentiation yields the density function of the range: given in this case by

$$
f_R(a) = \begin{cases} n(n-1)a^{n-2}(1-a) & 0 \le a \le 1 \\ 0 & \text{otherwise} \end{cases}
$$

That is, the range of n independent uniform (0, 1) random variables is a beta random variable with parameters $n - 1, 2$. ■

6.7 JOINT PROBABILITY DISTRIBUTION OF FUNCTIONS OF RANDOM VARIABLES

Let X_1 and X_2 be jointly continuous random variables with joint probability density function f_{X_1,X_2}. It is sometimes necessary to obtain the joint distribution of the random variables Y_1 and Y_2, which arise as functions of X_1 and X_2. Specifically, suppose that $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ for some functions g_1 and g_2.

Assume that the functions g_1 and g_2 satisfy the following conditions:

1. The equations $y_1 = g_1(x_1, x_2)$ and $y_2 = g_2(x_1, x_2)$ can be uniquely solved for x_1 and x_2 in terms of y_1 and y_2, with solutions given by, say, $x_1 = h_1(y_1, y_2), x_2 = h_2(y_1, y_2)$.
2. The functions g_1 and g_2 have continuous partial derivatives at all points (x_1, x_2) and are such that the 2×2 determinant

$$J(x_1, x_2) = \begin{vmatrix} \dfrac{\partial g_1}{\partial x_1} & \dfrac{\partial g_1}{\partial x_2} \\ \dfrac{\partial g_2}{\partial x_1} & \dfrac{\partial g_2}{\partial x_2} \end{vmatrix} \equiv \frac{\partial g_1}{\partial x_1}\frac{\partial g_2}{\partial x_2} - \frac{\partial g_1}{\partial x_2}\frac{\partial g_2}{\partial x_1} \neq 0$$

at all points (x_1, x_2).

Under these two conditions, it can be shown that the random variables Y_1 and Y_2 are jointly continuous with joint density function given by

$$f_{Y_1Y_2}(y_1, y_2) = f_{X_1,X_2}(x_1, x_2)|J(x_1, x_2)|^{-1} \tag{7.1}$$

where $x_1 = h_1(y_1,\ y_2),\ x_2 = h_2(y_1, y_2)$.

A proof of Equation (7.1) would proceed along the following lines:

$$P\{Y_1 \leq y_1, Y_2 \leq y_2\} = \iint\limits_{\substack{(x_1,x_2):\\ g_1(x_1,x_2) \leq y_1\\ g_2(x_1,x_2) \leq y_2}} f_{X_1,X_2}(x_1, x_2)\, dx_1\, dx_2 \tag{7.2}$$

The joint density function can now be obtained by differentiating Equation (7.2) with respect to y_1 and y_2. That the result of this differentiation will be equal to the right-hand side of Equation (7.1) is an exercise in advanced calculus whose proof will not be presented in this book.

EXAMPLE 7a

Let X_1 and X_2 be jointly continuous random variables with probability density function f_{X_1,X_2}. Let $Y_1 = X_1 + X_2, Y_2 = X_1 - X_2$. Find the joint density function of Y_1 and Y_2 in terms of f_{X_1,X_2}.

Solution. Let $g_1(x_1, x_2) = x_1 + x_2$ and $g_2(x_1, x_2) = x_1 - x_2$. Then

$$J(x_1, x_2) = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$$

Also, since the equations $y_1 = x_1 + x_2$ and $y_2 = x_1 - x_2$ have $x_1 = (y_1 + y_2)/2$, $x_2 = (y_1 - y_2)/2$ as their solution, it follows from Equation (7.1) that the desired density is

$$f_{Y_1,Y_2}(y_1, y_2) = \frac{1}{2} f_{X_1,X_2}\left(\frac{y_1 + y_2}{2}, \frac{y_1 - y_2}{2}\right)$$

For instance, if X_1 and X_2 are independent uniform (0, 1) random variables, then

$$f_{Y_1,Y_2}(y_1, y_2) = \begin{cases} \frac{1}{2} & 0 \leq y_1 + y_2 \leq 2,\ 0 \leq y_1 - y_2 \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

or if X_1 and X_2 are independent exponential random variables with respective parameters λ_1 and λ_2, then

$$f_{Y_1,Y_2}(y_1,y_2) = \begin{cases} \dfrac{\lambda_1\lambda_2}{2} \exp\left\{-\lambda_1\left(\dfrac{y_1+y_2}{2}\right) - \lambda_2\left(\dfrac{y_1-y_2}{2}\right)\right\} & y_1+y_2 \geq 0,\ y_1-y_2 \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

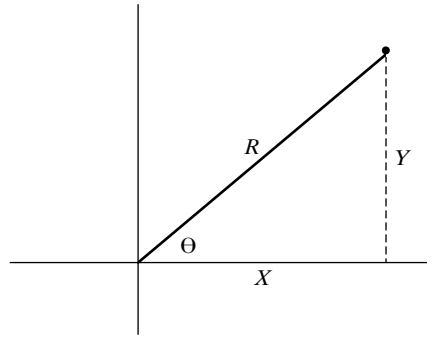


FIGURE 6.4: • = Random point. $(X, Y) = (R, \Theta)$.

Finally, if X_1 and X_2 are independent standard normal random variables, then

$$\begin{aligned} f_{Y_1,Y_2}(y_1,y_2) &= \frac{1}{4\pi} e^{-[(y_1+y_2)^2/8+(y_1-y_2)^2/8]} \\ &= \frac{1}{4\pi} e^{-(y_1^2+y_2^2)/4} \\ &= \frac{1}{\sqrt{4\pi}} e^{-y_1^2/4} \frac{1}{\sqrt{4\pi}} e^{-y_2^2/4} \end{aligned}$$

Thus, not only do we obtain (in agreement with Proposition 3.2) that both $X_1 + X_2$ and $X_1 - X_2$ are normal with mean 0 and variance 2, but we also conclude that these two random variables are independent. (In fact, it can be shown that if X_1 and X_2 are independent random variables having a common distribution function F, then $X_1 + X_2$ will be independent of $X_1 - X_2$ if and only if F is a normal distribution function.) ■

EXAMPLE 7b

Let (X, Y) denote a random point in the plane, and assume that the rectangular coordinates X and Y are independent standard normal random variables. We are interested in the joint distribution of R, Θ, the polar coordinate representation of (x, y). (See Figure 6.4.)

Suppose first that X and Y are both positive. For x and y positive, letting $r = g_1(x,y) = \sqrt{x^2+y^2}$ and $\theta = g_2(x,y) = \tan^{-1} y/x$, we see that

$$\begin{aligned} \frac{\partial g_1}{\partial x} &= \frac{x}{\sqrt{x^2+y^2}} \\ \frac{\partial g_1}{\partial y} &= \frac{y}{\sqrt{x^2+y^2}} \end{aligned}$$

$$\frac{\partial g_2}{\partial x} = \frac{1}{1 + (y/x)^2}\left(\frac{-y}{x^2}\right) = \frac{-y}{x^2 + y^2}$$
$$\frac{\partial g_2}{\partial y} = \frac{1}{x[1 + (y/x)^2]} = \frac{x}{x^2 + y^2}$$

Hence,

$$J(x, y) = \frac{x^2}{(x^2 + y^2)^{3/2}} + \frac{y^2}{(x^2 + y^2)^{3/2}} = \frac{1}{\sqrt{x^2 + y^2}} = \frac{1}{r}$$

Because the conditional joint density function of X, Y given that they are both positive is

$$f(x, y|X > 0, Y > 0) = \frac{f(x, y)}{P(X > 0, Y > 0)} = \frac{2}{\pi} e^{-(x^2+y^2)/2}, \; x > 0, y > 0$$

we see that the conditional joint density function of $R = \sqrt{X^2 + Y^2}$ and $\Theta = \tan^{-1}(Y/X)$, given that X and Y are both positive, is

$$f(r, \theta|X > 0, Y > 0) = \frac{2}{\pi} re^{-r^2/2}, \quad 0 < \theta < \pi/2, \quad 0 < r < \infty$$

Similarly, we can show that

$$f(r, \theta|X < 0, Y > 0) = \frac{2}{\pi} re^{-r^2/2}, \quad \pi/2 < \theta < \pi, \quad 0 < r < \infty$$
$$f(r, \theta|X < 0, Y < 0) = \frac{2}{\pi} re^{-r^2/2}, \quad \pi < \theta < 3\pi/2, \quad 0 < r < \infty$$
$$f(r, \theta|X > 0, Y < 0) = \frac{2}{\pi} re^{-r^2/2}, \quad 3\pi/2 < \theta < 2\pi, \quad 0 < r < \infty$$

As the joint density is an equally weighted average of these 4 conditional joint densities, we obtain that the joint density of R, Θ is given by

$$f(r, \theta) = \frac{1}{2\pi} re^{-r^2/2} \qquad 0 < \theta < 2\pi, \qquad 0 < r < \infty$$

Now, this joint density factors into the marginal densities for R and Θ, so R and Θ are independent random variables, with Θ being uniformly distributed over $(0, 2\pi)$ and R having the Rayleigh distribution with density

$$f(r) = re^{-r^2/2} \qquad 0 < r < \infty$$

(For instance, when one is aiming at a target in the plane, if the horizontal and vertical miss distances are independent standard normals, then the absolute value of the error has the preceding Rayleigh distribution.)

This result is quite interesting, for it certainly is not evident a priori that a random vector whose coordinates are independent standard normal random variables will have an angle of orientation that not only is uniformly distributed, but also is independent of the vector's distance from the origin.

If we wanted the joint distribution of R^2 and Θ, then, since the transformation $d = g_1(x, y) = x^2 + y^2$ and $\theta = g_2(x, y) = \tan^{-1} y/x$ has the Jacobian

$$J = \begin{vmatrix} 2x & 2y \\ \dfrac{-y}{x^2 + y^2} & \dfrac{x}{x^2 + y^2} \end{vmatrix} = 2$$

it follows that

$$f(d, \theta) = \frac{1}{2} e^{-d/2} \frac{1}{2\pi} \qquad 0 < d < \infty, \qquad 0 < \theta < 2\pi$$

Therefore, R^2 and Θ are independent, with R^2 having an exponential distribution with parameter $\frac{1}{2}$. But because $R^2 = X^2 + Y^2$, it follows by definition that R^2 has a chi-squared distribution with 2 degrees of freedom. Hence, we have a verification of the result that the exponential distribution with parameter $\frac{1}{2}$ is the same as the chi-squared distribution with 2 degrees of freedom.

The preceding result can be used to simulate (or generate) normal random variables by making a suitable transformation on uniform random variables. Let U_1 and U_2 be independent random variables, each uniformly distributed over $(0, 1)$. We will transform U_1, U_2 into two independent unit normal random variables X_1 and X_2 by first considering the polar coordinate representation (R, Θ) of the random vector (X_1, X_2). From the preceding, R^2 and Θ will be independent, and, in addition, $R^2 = X_1^2 + X_2^2$ will have an exponential distribution with parameter $\lambda = \frac{1}{2}$. But $-2 \log U_1$ has such a distribution, since, for $x > 0$,

$$\begin{aligned} P\{-2 \log U_1 < x\} &= P\left\{\log U_1 > -\frac{x}{2}\right\} \\ &= P\{U_1 > e^{-x/2}\} \\ &= 1 - e^{-x/2} \end{aligned}$$

Also, because $2\pi U_2$ is a uniform $(0, 2\pi)$ random variable, we can use it to generate Θ. That is, if we let

$$\begin{aligned} R^2 &= -2 \log U_1 \\ \Theta &= 2\pi U_2 \end{aligned}$$

then R^2 can be taken to be the square of the distance from the origin and θ can be taken to be the angle of orientation of (X_1, X_2). Now, since $X_1 = R \cos \Theta, X_2 = R \sin \Theta$, it follows that

$$\begin{aligned} X_1 &= \sqrt{-2 \log U_1} \cos(2\pi U_2) \\ X_2 &= \sqrt{-2 \log U_1} \sin(2\pi U_2) \end{aligned}$$

are independent standard normal random variables. ■

EXAMPLE 7c

If X and Y are independent gamma random variables with parameters (α, λ) and (β, λ), respectively, compute the joint density of $U = X + Y$ and $V = X/(X + Y)$.

Solution. The joint density of X and Y is given by

$$\begin{aligned} f_{X,Y}(x,y) &= \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)} \frac{\lambda e^{-\lambda y}(\lambda y)^{\beta-1}}{\Gamma(\beta)} \\ &= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} e^{-\lambda(x+y)} x^{\alpha-1} y^{\beta-1} \end{aligned}$$

Now, if $g_1(x,y) = x + y, g_2(x,y) = x/(x+y)$, then

$$\frac{\partial g_1}{\partial x} = \frac{\partial g_1}{\partial y} = 1 \quad \frac{\partial g_2}{\partial x} = \frac{y}{(x+y)^2} \quad \frac{\partial g_2}{\partial y} = -\frac{x}{(x+y)^2}$$

so

$$J(x,y) = \begin{vmatrix} 1 & 1 \\ \dfrac{y}{(x+y)^2} & \dfrac{-x}{(x+y)^2} \end{vmatrix} = -\frac{1}{x+y}$$

Finally, as the equations $u = x + y, v = x/(x+y)$ have as their solutions $x = uv, y = u(1-v)$, we see that

$$\begin{aligned} f_{U,V}(u,v) &= f_{X,Y}[uv, u(1-v)]u \\ &= \frac{\lambda e^{-\lambda u}(\lambda u)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)} \frac{v^{\alpha-1}(1-v)^{\beta-1}\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \end{aligned}$$

Hence, $X + Y$ and $X/(X+Y)$ are independent, with $X + Y$ having a gamma distribution with parameters $(\alpha+\beta, \lambda)$ and $X/(X+Y)$ having a beta distribution with parameters (α, β). The preceding reasoning also shows that $B(\alpha,\beta)$, the normalizing factor in the beta density, is such that

$$\begin{aligned} B(\alpha,\beta) &\equiv \int_0^1 v^{\alpha-1}(1-v)^{\beta-1} dv \\ &= \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)} \end{aligned}$$

This entire result is quite interesting. For suppose there are $n + m$ jobs to be performed, each (independently) taking an exponential amount of time with rate λ to be completed and suppose that we have two workers to perform these jobs. Worker I will do jobs $1, 2, \ldots, n$, and worker II will do the remaining m jobs. If we let X and Y denote the total working times of workers I and II, respectively, then (either from the foregoing result or from Example 3b) X and Y will be independent gamma random variables having parameters (n, λ) and (m, λ), respectively. It then follows that, independently of the working time needed to complete all $n + m$ jobs (that is, of $X + Y$), the proportion of this work that will be performed by worker I has a beta distribution with parameters (n, m). ■

When the joint density function of the n random variables $X_1, X_2, \ldots, X_n$ is given and we want to compute the joint density function of $Y_1, Y_2, \ldots, Y_n$, where

$$Y_1 = g_1(X_1, \ldots, X_n) \quad Y_2 = g_2(X_1, \ldots, X_n), \ldots \quad Y_n = g_n(X_1, \ldots, X_n)$$

the approach is the same—namely, we assume that the functions g_i have continuous partial derivatives and that the Jacobian determinant.

$$J(x_1, \ldots, x_n) = \begin{vmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \cdots & \frac{\partial g_2}{\partial x_n} \\ \frac{\partial g_n}{\partial x_1} & \frac{\partial g_n}{\partial x_2} & \cdots & \frac{\partial g_n}{\partial x_n} \end{vmatrix} \neq 0$$

at all points $(x_1, \ldots, x_n)$. Furthermore, we suppose that the equations $y_1 = g_1(x_1, \ldots, x_n), y_2 = g_2(x_1, \ldots, x_n), \ldots, y_n = g_n(x_1, \ldots, x_n)$ have a unique solution, say, $x_1 = h_1(y_1, \ldots, y_n), \ldots, x_n = h_n(y_1, \ldots, y_n)$. Under these assumptions, the joint density function of the random variables Y_i is given by

$$f_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n) = f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)|J(x_1, \ldots, x_n)|^{-1} \tag{7.3}$$

where $x_i = h_i(y_1, \ldots, y_n), i = 1, 2, \ldots, n$.

EXAMPLE 7d

Let X_1, X_2, and X_3 be independent standard normal random variables. If $Y_1 = X_1 + X_2 + X_3, Y_2 = X_1 - X_2$, and $Y_3 = X_1 - X_3$, compute the joint density function of Y_1, Y_2, Y_3.

Solution. Letting $Y_1 = X_1 + X_2 + X_3, Y_2 = X_1 - X_2, Y_3 = X_1 - X_3$, the Jacobian of these transformations is given by

$$J = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{vmatrix} = 3$$

As the preceding transformations yield that

$$X_1 = \frac{Y_1 + Y_2 + Y_3}{3} \quad X_2 = \frac{Y_1 - 2Y_2 + Y_3}{3} \quad X_3 = \frac{Y_1 + Y_2 - 2Y_3}{3}$$

we see from Equation (7.3) that

$$f_{Y_1, Y_2, Y_3}(y_1, y_2, y_3) = \frac{1}{3} f_{X_1, X_2, X_3}\left(\frac{y_1 + y_2 + y_3}{3}, \frac{y_1 - 2y_2 + y_3}{3}, \frac{y_1 + y_2 - 2y_3}{3}\right)$$

Hence, as

$$f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \frac{1}{(2\pi)^{3/2}} e^{-\sum_{i=1}^{3} x_i^2/2}$$

we see that

$$f_{Y_1, Y_2, Y_3}(y_1, y_2, y_3) = \frac{1}{3(2\pi)^{3/2}} e^{-Q(y_1, y_2, y_3)/2}$$

where

$$
\begin{aligned}
&Q(y_1, y_2, y_3) \\
&\quad = \left(\frac{y_1 + y_2 + y_3}{3}\right)^2 + \left(\frac{y_1 - 2y_2 + y_3}{3}\right)^2 + \left(\frac{y_1 + y_2 - 2y_3}{3}\right)^2 \\
&\quad = \frac{y_1^2}{3} + \frac{2}{3}y_2^2 + \frac{2}{3}y_3^2 - \frac{2}{3}y_2y_3
\end{aligned}
$$

■

EXAMPLE 7e

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed exponential random variables with rate λ. Let

$$Y_i = X_1 + \cdots + X_i \quad i = 1, \ldots, n$$

(a) Find the joint density function of $Y_1, \ldots, Y_n$.
(b) Use the result of part (a) to find the density of Y_n.

Solution. (a) The Jacobian of the transformations $Y_1 = X_1$, $Y_2 = X_1 + X_2$, ..., $Y_n = X_1 + \cdots + X_n$ is

$$
J = \begin{vmatrix}
1 & 0 & 0 & 0 & \cdots & 0 \\
1 & 1 & 0 & 0 & \cdots & 0 \\
1 & 1 & 1 & 0 & \cdots & 0 \\
\cdots & & \cdots & & & \\
\cdots & & \cdots & & & \\
1 & 1 & 1 & 1 & \cdots & 1
\end{vmatrix}
$$

Since only the first term of the determinant will be nonzero, we have $J = 1$. Now, the joint density function of $X_1, \ldots, X_n$ is given by

$$f_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} \quad 0 < x_i < \infty,\ i = 1, \ldots, n$$

Hence, because the preceding transformations yield

$$X_1 = Y_1, X_2 = Y_2 - Y_1, \ldots, X_i = Y_i - Y_{i-1}, \ldots, X_n = Y_n - Y_{n-1}$$

it follows from Equation (7.3) that the joint density function of $Y_1, \ldots, Y_n$ is $f_{Y_1,\ldots,Y_n}(y_1, y_2, \ldots, y_n)$

$$
\begin{aligned}
&= f_{X_1,\ldots,X_n}(y_1, y_2 - y_1, \ldots, y_i - y_{i-1}, \ldots, y_n - y_{n-1}) \\
&= \lambda^n \exp\left\{-\lambda\left[y_1 + \sum_{i=2}^{n}(y_i - y_{i-1})\right]\right\} \\
&= \lambda^n e^{-\lambda y_n} \quad 0 < y_1, 0 < y_i - y_{i-1}, i = 2, \ldots, n \\
&= \lambda^n e^{-\lambda y_n} \quad 0 < y_1 < y_2 < \cdots < y_n
\end{aligned}
$$

(b) To obtain the marginal density of Y_n, let us integrate out the other variables one at a time. Doing this gives

$$f_{Y_2,\ldots,Y_n}(y_2,\ldots,y_n) = \int_0^{y_2} \lambda^n e^{-\lambda y_n} dy_1$$
$$= \lambda^n y_2 e^{-\lambda y_n} \quad 0 < y_2 < y_3 < \cdots < y_n$$

Continuing, we obtain

$$f_{Y_3,\ldots,Y_n}(y_3,\ldots,y_n) = \int_0^{y_3} \lambda^n y_2 e^{-\lambda y_n} dy_2$$
$$= \lambda^n \frac{y_3^2}{2} e^{-\lambda y_n} \quad 0 < y_3 < y_4 < \cdots < y_n$$

The next integration yields

$$f_{Y_4,\ldots,Y_n}(y_4,\ldots,y_n) = \lambda^n \frac{y_4^3}{3!} e^{-\lambda y_n} \quad 0 < y_4 < \cdots < y_n$$

Continuing in this fashion gives

$$f_{Y_n}(y_n) = \lambda^n \frac{y_n^{n-1}}{(n-1)!} e^{-\lambda y_n} \quad 0 < y_n$$

which, in agreement with the result obtained in Example 3b, shows that $X_1 + \cdots + X_n$ is a gamma random variable with parameters n and λ. ■

*6.8 EXCHANGEABLE RANDOM VARIABLES

The random variables $X_1, X_2, \ldots, X_n$ are said to be *exchangeable* if, for every permutation $i_1, \ldots, i_n$ of the integers $1, \ldots, n$,

$$P\{X_{i_1} \le x_1, X_{i_2} \le x_2, \ldots, X_{i_n} \le x_n\} = P\{X_1 \le x_1, X_2 \le x_2, \ldots, X_n \le x_n\}$$

for all $x_1, \ldots, x_n$. That is, the n random variables are exchangeable if their joint distribution is the same no matter in which order the variables are observed.

Discrete random variables will be exchangeable if

$$P\{X_{i_1} = x_1, X_{i_2} = x_2, \ldots, X_{i_n} = x_n\} = P\{X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n\}$$

for all permutations $i_1, \ldots, i_n$, and all values $x_1, \ldots, x_n$. This is equivalent to stating that $p(x_1, x_2, \ldots, x_n) = P\{X_1 = x_1, \ldots, X_n = x_n\}$ is a symmetric function of the vector $(x_1, \ldots, x_n)$, which means that its value does not change when the values of the vector are permuted.

EXAMPLE 8a

Suppose that balls are withdrawn one at a time and without replacement from an urn that initially contains n balls, of which k are considered special, in such a manner that each withdrawal is equally likely to be any of the balls that remain in the urn at the time. Let $X_i = 1$ if the ith ball withdrawn is special and let $X_i = 0$ otherwise. We will show that the random variables $X_1, \ldots, X_n$ are exchangeable. To do so, let $(x_1, \ldots, x_n)$ be a vector consisting of k ones and $n - k$ zeros. However, before considering the joint mass function evaluated at $(x_1, \ldots, x_n)$, let us try to gain some insight by